

2.4

Error Analysis for Iterative Methods

1. Order of Convergence of a Sequence

► Definition: Suppose the sequence

$$p_0, p_1, p_2, \dots, p_n, \dots \overset{\text{approaches}}{\boxed{->}} p \leftarrow \text{A constant}$$

If $\lambda > 0$ and $\alpha > 0$ exist with

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} \boxed{-} p|}{|p_n \boxed{-} p|^\alpha} = \lambda$$

then the sequence $p_0, p_1, \dots, p_n, \dots$ converges to p of order α .

Order of Convergence

► **Definition:** λ is called **asymptotic error constant**.

► Important cases:

- If $\alpha = 1$ and $0 < \lambda \leq 1$, then the sequence is linearly convergent.
- If $\alpha = 2$, then the sequence is quadratically convergent.

► Example: 1. Show that the sequence converges to 0 linearly.

$p=0$

$$\left| \frac{\frac{1}{(n+1)^2} - 0}{\frac{1}{n^2} - 0} \right| = \frac{n^2}{(n+1)^2} = \frac{1}{n^2 + 2n + 1} \rightarrow 1$$

$p=0$

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)^2} - 0}{\frac{1}{n^2} - 0} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^2}{(n+1)^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^2}{n^2 + 2n + 1} \right|$$

$$\frac{1}{n+1} \times n^2$$

$$\left(\frac{1}{\infty} = 0 \right)$$

$$\frac{2}{\infty} = 0 \quad \frac{1}{\infty} = \frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{1 + \frac{2}{n} + \frac{1}{n^2}} = \frac{1}{1 + 0 + 0} = 1$$

$$P_n = \frac{1}{n^2} \quad P=0 \quad \alpha=1$$

$$\lim_{n \rightarrow \infty} \frac{|P_{n+1} - P|}{|P_n - P|^\alpha} = \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)^2} - 0 \right|}{\left| \frac{1}{n^2} - 0 \right|^1} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\frac{n}{x}}{\frac{n}{x} + \frac{1}{x}} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)$$

$$= \frac{1}{1 + \frac{1}{\infty}} = \frac{1}{1} = 1 \quad \text{Since it equals 1, } P_n \rightarrow P \text{ linearly converges}$$

$P_n = \frac{1}{n^2}$, does this sequence converges linearly?

$$\lim_{n \rightarrow \infty} \frac{|P_{n+1} - P|}{|P_n - P|^\alpha} = \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)^2} - 0 \right|}{\left| \frac{1}{n^2} - 0 \right|^1} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \left| \frac{n^2}{(n+1)^2} \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1} \right)^2 \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{\frac{n}{x}}{\frac{n}{x} + \frac{1}{x}} \right)^2 \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{1}{1 + \frac{1}{n}} \right)^2 \right| = \left| \left(\frac{1}{1} \right)^2 \right| = 1 \text{ so converges}$$

2. Show that the sequence $p_n = 10^{-2^n}$ converges to zero quadratically.

$$p = 0$$

$$p_{n+1} = 10^{-2^{n+1}}$$

$$10^{-2^n} = \frac{1}{10^{2^n}} \quad \begin{matrix} \text{so} \\ \text{pos} \\ \text{num} \end{matrix}$$

$$\frac{|p_{n+1} - 0|}{|p_n - 0|^\alpha} = \frac{10^{-2^{n+1}}}{10^{-2 \cdot 2^n}} = \frac{10^{-2 \cdot 2^n}}{10^{-2 \cdot 2^n}} = 1$$

$$\lim_{n \rightarrow \infty} \frac{|10^{-2^{n+1}} - 0|}{|10^{-2^n} - 0|^2} = \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{(10^{-2^n})^2} = \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{10^{-2^n \cdot 2}} = \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{10^{-2^{n+1}}} = 1$$

p_n converges to p quadratically

2. Show that the sequence to zero quadratically.

$$p_{n+1} = 10^{-2^{n+1}} \quad p_n = 10^{-2^n}$$

$$p = 0$$

$$\alpha = 2$$

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lim_{n \rightarrow \infty} \frac{|10^{-2^{n+1}} - 0|}{|10^{-2^n} - 0|^2} = \lim_{n \rightarrow \infty} \frac{|10^{-2^{n+1}}|}{|10^{-2^{n+1}}|} = 1$$

$$(10^{-2^n})^2$$

$$10^{-2^{n+1}} = 10^{-2^{n+1}}$$

$$10^{-4^n}$$

Is the sequence $p_n = \frac{1}{n^3}$ converges to $\frac{0}{p=0}$
 quadratically?
 $\alpha = 2$

$$p_{n+1} = \frac{1}{(n+1)^3}$$

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha}$$

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)^3} - 0 \right|}{\left| \frac{1}{n^3} - 0 \right|^2} = \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)^3} \right|}{\left| \left(\frac{1}{n^3} \right)^2 \right|}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{n^6}{n^3}}{\frac{n^6}{n^3} + \frac{1}{n^3}}$$

divide by the highest power of the denominator

$$= \lim_{n \rightarrow \infty} \left| n^3 \right| = \infty$$

Does not converge quadratically.
 however it converges linearly)

$$= \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{(n+1)^3} \right|}{\left| \frac{1}{n^6} \right|} = \lim_{n \rightarrow \infty} \frac{n^6}{(n+1)^3}$$

$$\frac{n^6}{n^6 + 1} = \frac{n^3}{1 + \frac{1}{n^3}} = \frac{n^3}{1} = n^3$$

both have power 3

2.4 Exercise-7(a)

Examples:

1. Let $x_n = \frac{1}{n^k}$ for some fixed $k > 0$. Then we know this sequence converges to 0. Compute the rate of convergence:

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|^\alpha} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)^k}}{\frac{1}{n^{\alpha k}}} = \lim_{n \rightarrow \infty} \left(\frac{n^\alpha}{n+1} \right)^k$$

We get convergence for $\alpha = 1$, but not $\alpha = 2$, so this sequence converges **linearly** to 0.

This is interesting- Sequences like the following:

$$x_n = \frac{1}{n}, \quad x_n = \frac{1}{n^2}, \quad x_n = \frac{1}{n^3}, \dots$$

all converge **linearly** to zero.

2.4 Exercise-8(a)

2. Let $x_n = 10^{-2^n}$ (which converges to 0). Compute the rate of convergence:

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|^\alpha} = \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{10^{-2^n \cdot \alpha}}$$

It's easy to see that if $\alpha = 2$, we get convergence.

2.4 Exercise-8(b)

Show that the sequence $p_n = 10^{-n^k}$ does not converge quadratically, regardless of the size of the exponent k .

Since

$$\frac{|p_{n+1} - 0|}{|p_n - 0|^2} = \frac{10^{-(n+1)^k}}{10^{-2n^k}} = 10^{2n^k - (n+1)^k} \rightarrow \infty$$

as $n \rightarrow \infty$, we have that p_n does not converge quadratically to 0, for any positive integer k .

Exercise 2.4

1. Use Newton's method to find solutions accurate to within 10^{-5} to the following problems.
 - a. $x^2 - 2xe^{-x} + e^{-2x} = 0$, for $0 \leq x \leq 1$

Modified Newton's Method

Newton's method: $x_{n+1} = x_n - \frac{f(x)}{f'(x)}$
or $p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$ ↖ The same

Given an initial approximation p_0 , generate a sequence $\{p_n\}_{n=0}^{\infty}$ by

Modified
Newton's
method
(will be given
in the test)

$$p_n = p_{n-1} - \frac{f(p_{n-1})f'(p_{n-1})}{[f'(p_{n-1})]^2 - f(p_{n-1})f''(p_{n-1})}, \quad \text{for } n \geq 1.$$

Note: The modified Newton' method requires the second-order derivative $f''(x)$.

- Use Modified Newton's method to find solutions accurate to within 10^{-2} to the following problem.

- $2x \cos 2x - (x-2)^2 = 0$, for $[3, 4]$ $P_0 = 3.5$

$$f(x) = 2x \cos 2x - (x-2)^2 = 0$$

$$f'(x) = 2 \cos 2x + 2x(-\sin 2x) \cdot 2 - 2(x-2) = 2 \cos 2x - 4x \sin 2x - 2x + 4$$

$$f''(x) = 2(-\sin 2x) \cdot 2 - 4 \sin 2x - 4x(\cos 2x) \cdot 2 - 2 = -4 \sin 2x - 4 \sin 2x - 8x \cos 2x - 2 = -8 \sin 2x - 8 \cos 2x - 2$$

$$P_n = P_{n-1} - \frac{f(P_{n-1}) f'(P_{n-1})}{[f'(P_{n-1})]^2 - f(P_{n-1}) f''(P_{n-1})} \quad P_1 = 3.5 - \frac{(-3.02715)(-10.690007)}{(-10.690007)^2 - (-3.02715)(-28.365156)} = 3.661692$$

$$f(P_0) = f(3.5) = 2(3.5) \cos(2(3.5)) - (3.5-2)^2 = 3.027315$$

$$f'(3.5) = 2 \cos(2(3.5)) - 4(3.5) \sin(2(3.5)) - 2(3.5) + 4 = -10.690007$$

$$f''(3.5) = -8 \sin(2(3.5)) - 8(3.5) \cos(2(3.5)) - 2 = -28.365156$$

$$p_1 = 3.661692$$

$$f(p_1) = 2(p_1) \cos(2(p_1)) - (p_1 - 2)^2 =$$

$$f'(p_1) = 2 \cos(2(p_1)) - 4(p_1) \sin(2(p_1)) - 2(p_1) + 4 =$$

$$f''(p_1) = -8 \sin(2(p_1)) - 8(p_1) \cos(2(p_1)) - 2 =$$